

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Concept Mapping (Medium)

Assessment Tool Weightage: 20%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

Code of Conduct:

I SUBASHREE.N(192521377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

SUBASHREE.N

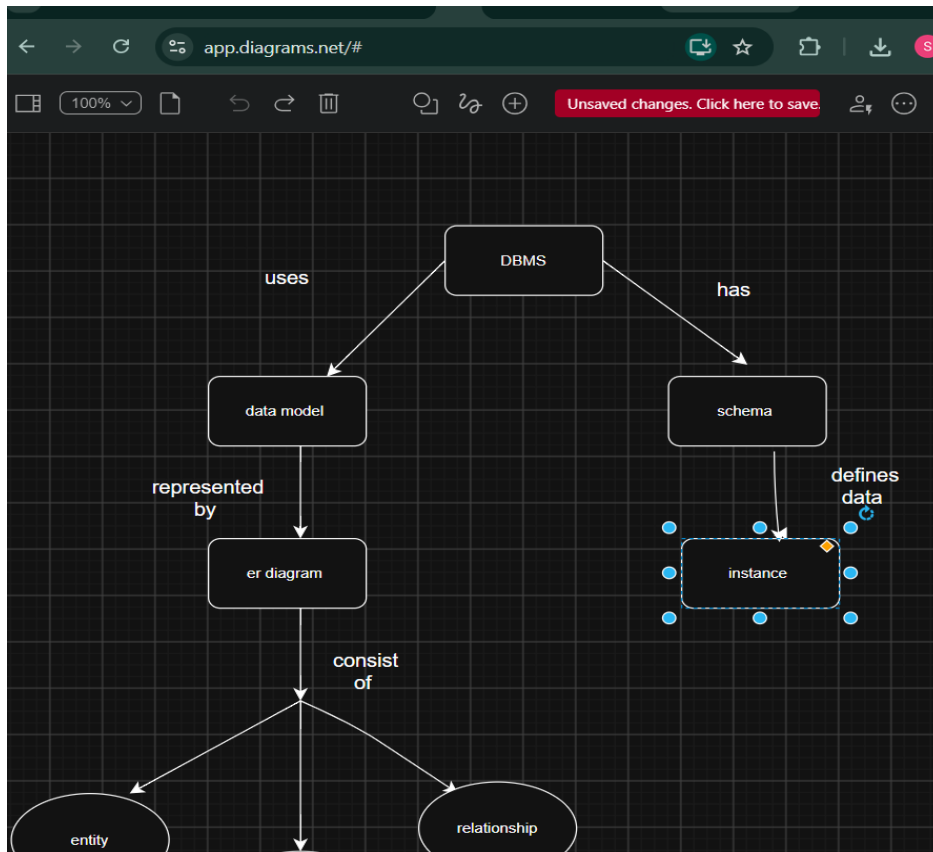
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

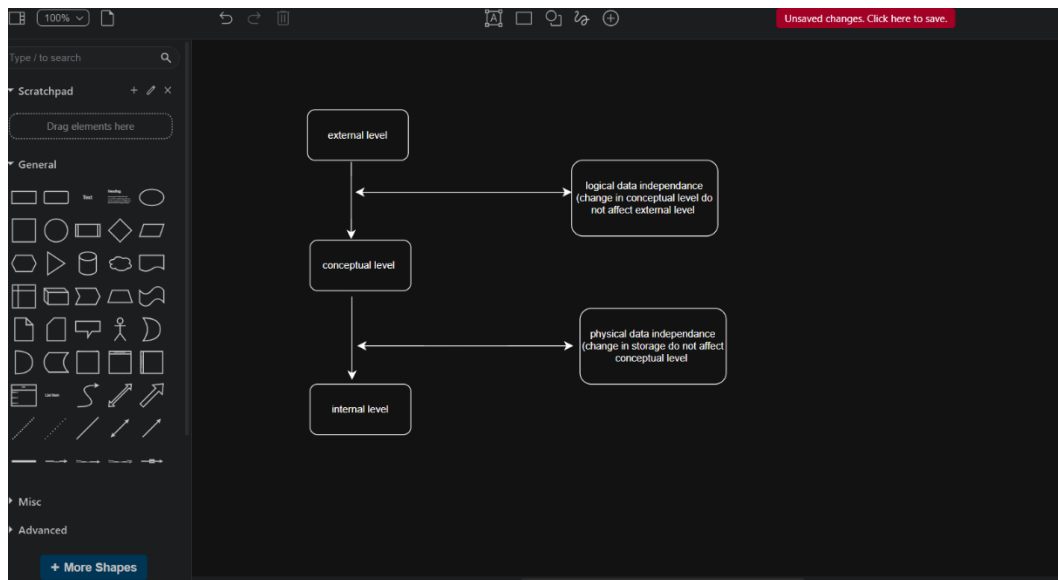
ANSWER 1:



Explanation

- **DBMS** manages and controls the database.
- **Data Model** defines how data is organized.
- **ER Diagram** represents the data model visually.
- **ER Diagram** consists of **Entity, Attribute, and Relationship**.
- **Schema** defines the structure of the database.
- **Instance** is the actual data stored in the database at a particular time.

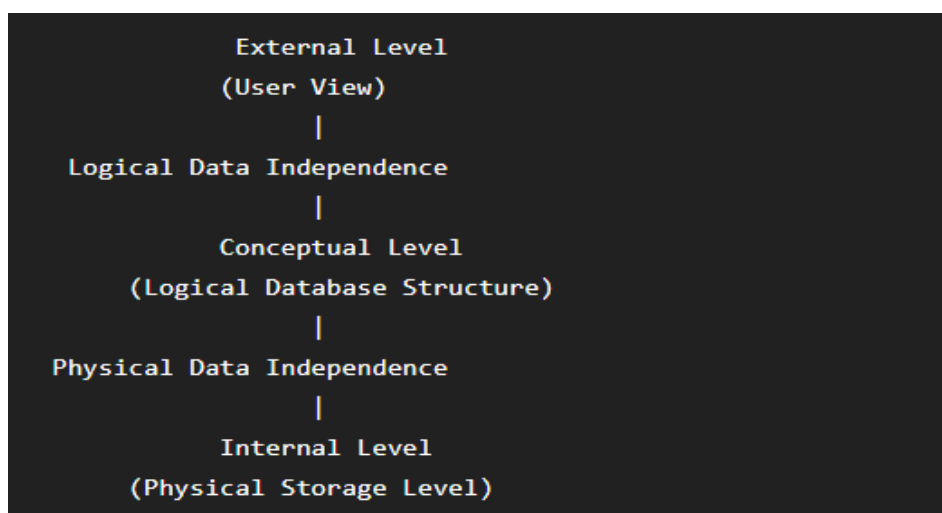
ANSWER 2:



Explanation

- **External Level** → User's view.
- **Conceptual Level** → Complete database structure.
- **Internal Level** → Physical storage.
- **Logical Data Independence** → Changes in conceptual level do not affect external level.
- **Physical Data Independence** → Changes in storage do not affect conceptual level.

ANSWER 3:

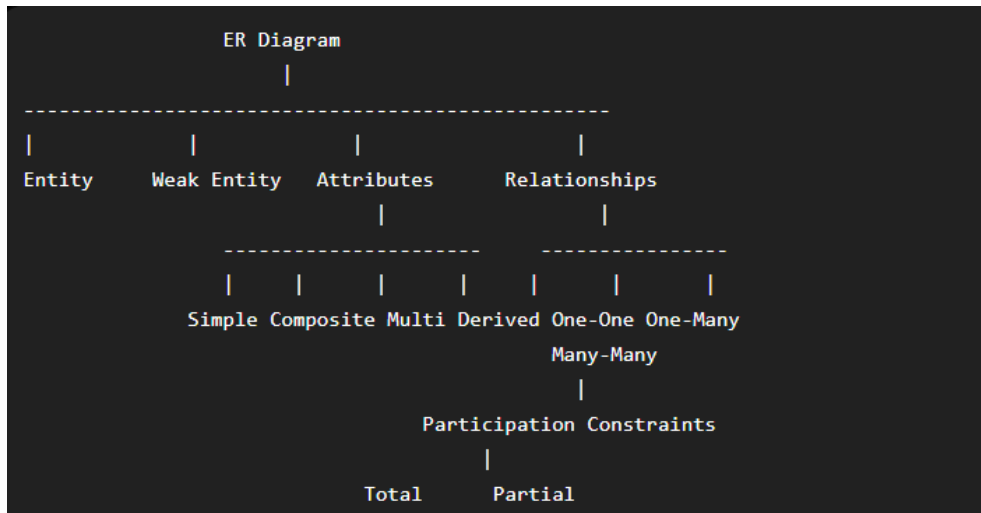


Problems solved by DBMS

- Removes duplicate data.

- Improves security.
- Provides backup and recovery.
- Supports multiple users.
- Maintains data consistency.

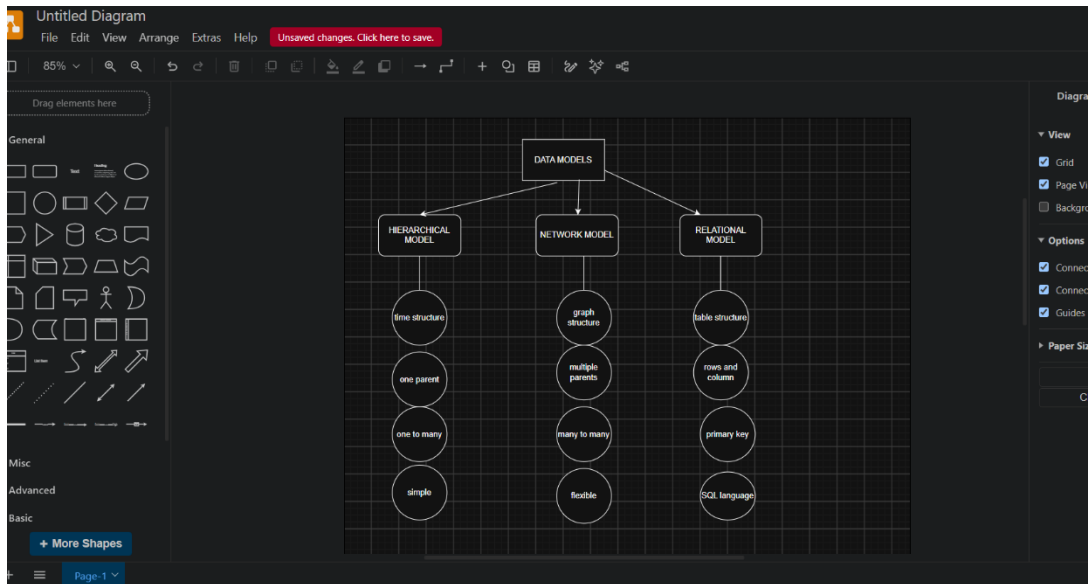
ANSWER 4:



Explanation

- **Entity** → Strong object.
- **Weak Entity** → Depends on another entity.
- **Attributes** → Simple, Composite, Multivalued, Derived.
- **Relationships** → 1:1, 1:M, M:N.
- **Participation** → Total or Partial.

ANSWER 5:



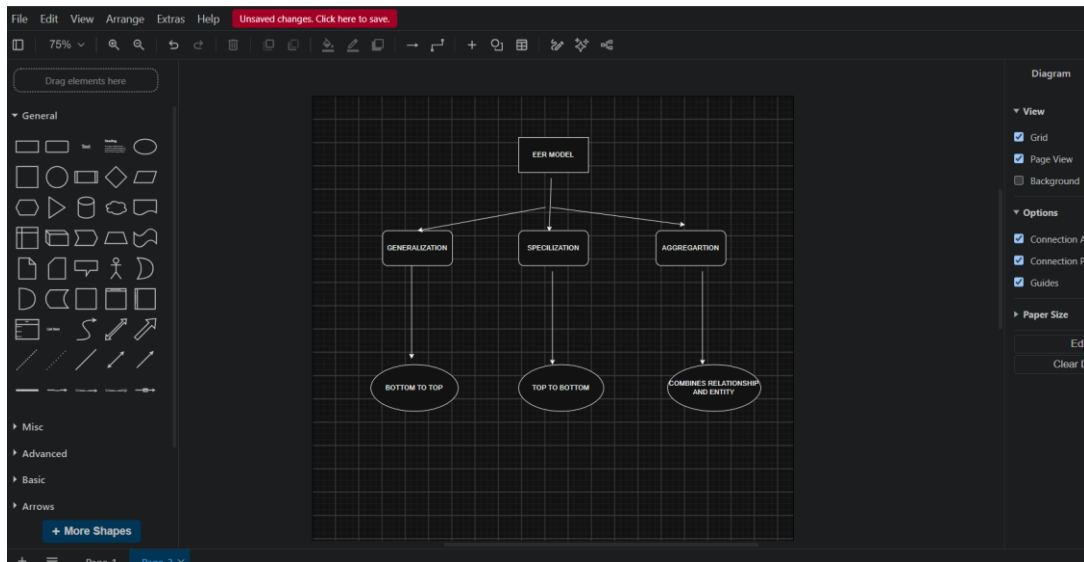
EXPLANATION:

- **Hierarchical Model** organizes data in a **tree structure** with **one-to-many relationships**.
- **Network Model** organizes data in a **graph structure** with **many-to-many relationships**.
- **Relational Model** stores data in **tables (rows and columns)** and uses **SQL** for data manipulation.

Comparison Table:

Hierarchical	Network	Relational
Tree structure	Graph structure	Table structure
One parent	Multiple parents	Tables
One to many	Many to many	Keys and sql

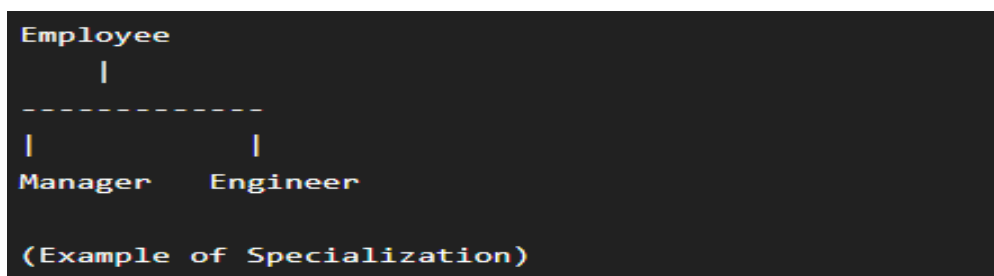
ANSWER 6:



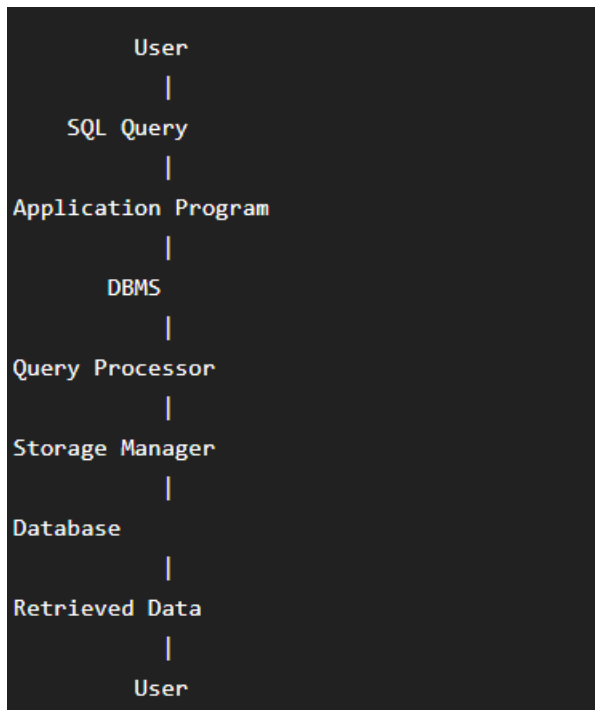
EXPLANATION:

- **EER Model** is an extension of the basic ER Model.
- **Generalization** combines similar lower-level entities into one higher-level entity.
- **Specialization** divides one higher-level entity into specialized subclasses.
- **Aggregation** treats a relationship and its participating entities as a higher-level entity for representing complex relationships.

EXAMPLE:



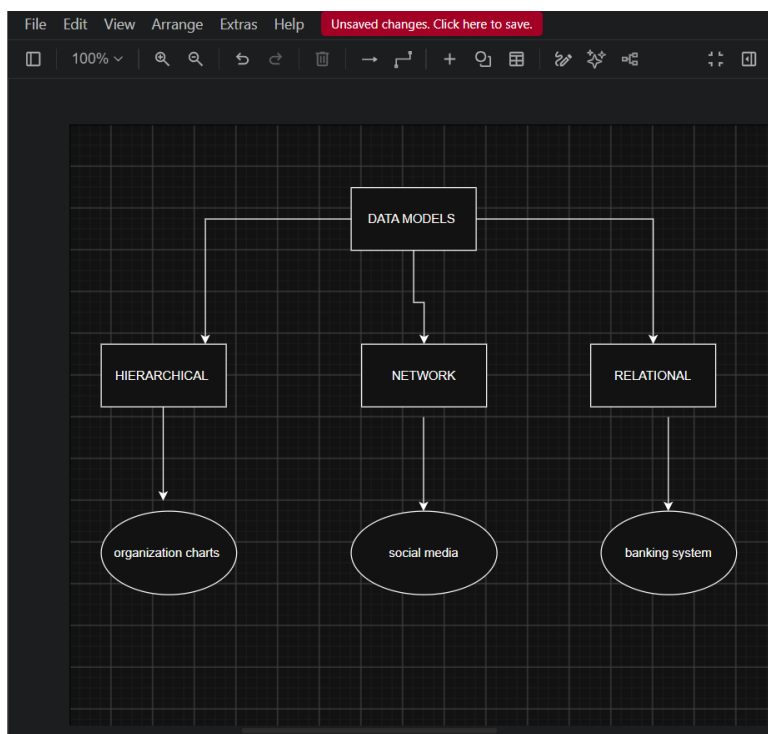
ANSWER 7:



Flow

User → Query → DBMS → Database → Result → User

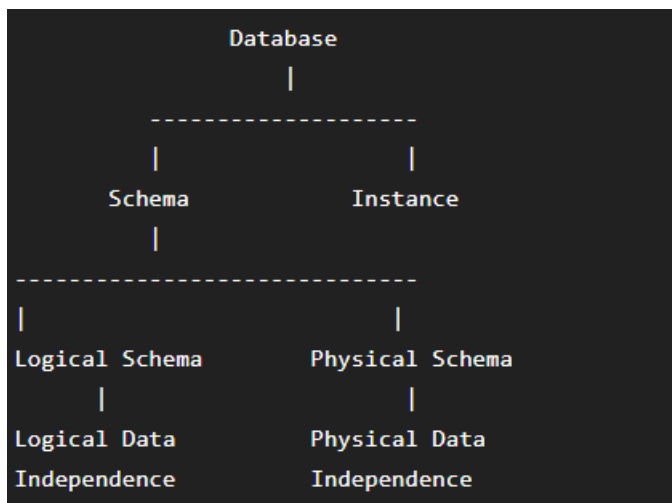
ANSWER 8:



EXPLANATION:

- **Hierarchical Model** is suitable for **organization charts** and **file systems** because it follows a parent-child hierarchy.
- **Network Model** is suitable for **social networks**, **telecommunication systems**, and **airline reservation systems** because it supports many-to-many relationships.
- **Relational Model** is suitable for **banking systems**, **hospital management systems**, and **college management systems** because it stores data in tables and supports SQL

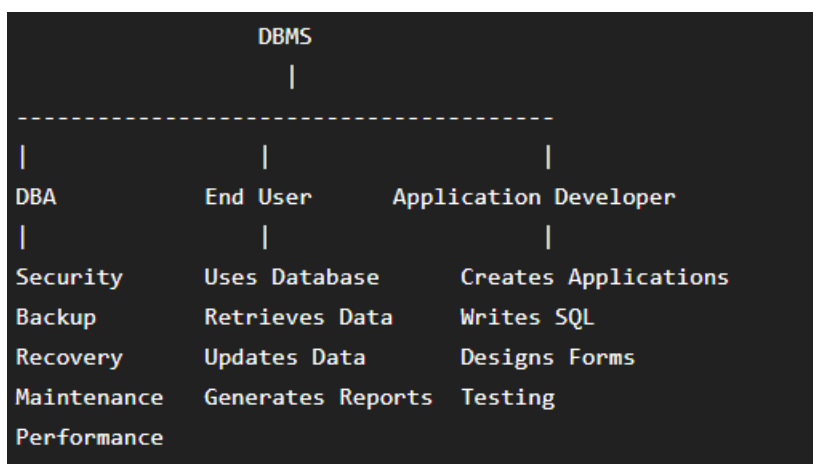
ANSWER 9:



Explanation

- **Schema** → Database structure.
- **Instance** → Actual data.
- **Logical Data Independence** → Changes in logical schema do not affect user views.
- **Physical Data Independence** → Changes in storage do not affect logical schema.

ANSWER 10:



Responsibilities

DBA

- Manages database.
- Provides security.
- Takes backup.
- Recovers data.
- Improves performance.

End User

- Inserts data.
- Updates data.
- Deletes data.
- Retrieves reports.

Application Developer

- Develops applications.
- Writes SQL queries.
- Designs forms.
- Tests applications.